CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 19: Mean and Median

Topic-wise short notes with solved examples

The arithmetic mean and the median of ungrouped data and of simple frequency distributions, the properties of the mean, and problems on corrected and combined means.

1 Arithmetic mean

- The **mean** $\bar{x}$ of n values $x_1, x_2, \ldots, x_n$ is

$$\bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{\sum x}{n}$$

- So $\sum x = n\bar{x}$: the sum of the values is the mean times their number.
- The sum of the **deviations** of the values from their mean is zero: $\sum(x - \bar{x}) = 0$.

Example 1.1 Find the mean of 12, 15, 9, 20, 14.

SOLUTION

$\frac{70}{5}$

14

Example 1.2 The mean of 5 numbers is 18. Four of them are 12, 20, 15 and 25. Find the fifth.

SOLUTION

Sum $= 90$; the four known numbers add up to 72.

18

2 Properties of the mean

- If every value is **increased** (or decreased) by a, the mean increases (or decreases) by a.
- If every value is **multiplied** by a, the mean is multiplied by a. If every value is **divided** by a ($a \neq 0$), the mean is divided by a.

Example 2.1 The mean of 10 numbers is 25. Find the new mean if each number is (i) increased by 4 (ii) multiplied by 3 (iii) divided by 5.

SOLUTION

(i) 29 (ii) 75 (iii) 5

Example 2.2 The mean of 8 values is 30. If each value is doubled and then 5 is subtracted, find the new mean.

SOLUTION

$2 \times 30 - 5$

55

3 Corrected and combined means

- **Wrong entry**: correct sum = recorded sum − wrong value + correct value. Then divide by n.
- **Value added or removed**: adjust the sum and the number of values.
- **Combined mean** of two groups with n_1, n_2 values and means $\bar{x}_1$, $\bar{x}_2$:

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

Example 3.1 The mean of 20 observations is 32. Later it is found that 46 was wrongly read as 64. Find the correct mean.

SOLUTION

Recorded sum $= 640$. Correct sum $= 640 - 64 + 46 = 622$.

31.1

Example 3.2 The mean weight of 24 students is 40 kg. When the teacher's weight is included, the mean rises by 1 kg. Find the teacher's weight.

SOLUTION

New sum $= 25 \times 41 = 1025$; old sum $= 960$.

65 kg

Example 3.3 The mean of 30 boys is 62 marks and the mean of 20 girls is 67 marks. Find the mean of the whole class.

SOLUTION

$$\frac{30 \times 62 + 20 \times 67}{50} = \frac{3200}{50}$$

64

Example 3.4 The mean of 9 numbers is 15. One number is removed and the mean becomes 14. Find the number removed.

SOLUTION

$135 - 8 \times 14$

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4 Mean of a frequency distribution

- If the values $x_1, x_2, \ldots$ occur with frequencies $f_1, f_2, \ldots$:

$$\bar{x} = \frac{\sum fx}{\sum f}$$

x	5	10	15	20	25	Total
f	4	6	10	7	3	30
fx	20	60	150	140	75	445

Example 4.1 Find the mean of the distribution in the table.

SOLUTION

$$\frac{445}{30}$$

14.83 (to two decimal places)

Example 4.2 The mean of the distribution below is 6. Find p. Values 2, 4, 6, 10, $p + 5$ with frequencies 3, 2, 3, 1, 2.

SOLUTION

$\sum f = 11$ and $\sum fx = 6 + 8 + 18 + 10 + 2p + 10 = 52 + 2p$.

$52 + 2p = 66$

$p = 7$

5 Median

- The **median** is the middle value when the data are arranged in ascending or descending order.
- For n values in order:

n	Median
Odd	The $\left(\frac{n+1}{2}\right)$th value
Even	The mean of the $\left(\frac{n}{2}\right)$th and $\left(\frac{n}{2}+1\right)$th values

- The median is not affected by very large or very small values at the ends. The mean is.

Example 5.1 Find the median of 17, 9, 24, 12, 30, 15, 21.

SOLUTION

In order: 9, 12, 15, 17, 21, 24, 30. The 4th value is the median.

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Example 5.2 Find the median of 8, 15, 3, 11, 20, 6, 13, 18.

SOLUTION

In order: 3, 6, 8, 11, 13, 15, 18, 20. The 4th and 5th values are 11 and 13.

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Example 5.3 The values $11, 13, 15, x+1, x+3, 30, 35, 40$ are in ascending order and their median is 22. Find x.

SOLUTION

The median is the mean of the 4th and 5th values: $\frac{(x+1)+(x+3)}{2} = x+2$.

$x = 20$

6 Median of a frequency distribution

- Write the cumulative frequencies. With total frequency N, find the position of the median as above, then read the value whose cumulative frequency first reaches that position.

x	3	5	7	9	11
f	4	8	10	6	3
Cumulative frequency	4	12	22	28	31

Example 6.1 Find the median of the distribution in the table.

SOLUTION

$N = 31$ is odd, so the median is the 16th value. Values from the 13th to the 22nd are all 7.

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7 Common mistakes

- Finding the median without arranging the data in order.
- For an even number of values, taking only one of the two middle values.
- Dividing $\sum fx$ by the number of different values instead of by $\sum f$.
- In a wrong-entry problem, subtracting the correct value and adding the wrong one.
- Forgetting to change the number of values when one is added or removed.
- Averaging two means directly when the groups have different sizes.